PROJECT REPORT
ON
" Advanced Persistent Threat(APT) Detection And Automated Incident Response Using Wazuh EDR and n8n "

Carried Out at

CENTRE FOR DEVELOPMENT OF ADVANCED COMPUTING ELECTRONIC CITY, BANGALORE.

UNDER THE SUPERVISION OF

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Submitted By

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PG DIPLOMA IN IT INFRASTRUCTURE, SYSTEMS & SECURITY C-DAC, BANGALORE

Candidate's Declaration

We hereby declare that the project work entitled "Advanced Persistent Threat (APT) Detection and Automated Incident Response Using Wazuh EDR and n8n" is an authentic record of our own work carried out as a requirement for the award of the degree of DIPLOMA in IT INFRASTRUCTURE SYSTEMS AND SECURITY at the CENTRE FOR DEVELOPMENT OF ADVANCED COMPUTING (C-DAC).

We certify that:

1. The matter embodied in this project report has not been submitted by us for the award of any other degree or diploma.

2. We have duly acknowledged all the sources, software tools, frameworks, and literature used in the preparation of this report.

3. The screenshots, logs, alerts, and experimental results presented in this report are based on the security monitoring and automation environment set up by us, comprising Wazuh EDR, n8n workflow automation, monitored endpoints (Windows/Linux), and supporting infrastructure, during the course of this project.

KULDEEP SINGH

Counter Signed by
Mr. Anshul Tailor

.....

ACKNOWLEDGMENT

I take this opportunity to express my gratitude to all those people who have been directly and indirectly with us during the competition of this project.

I pay thanks to MR. Anshul Tailor who has given guidance and a light to me during this major project. His versatile knowledge about "Advanced Persistent Threat(APT) Detection and Automated Incident Response using Wazuh EDR and n8n" has eased me in critical times during the span of this Final Project.

I acknowledge here our debt to those who contributed significantly to one or more steps. I take full responsibility for any remaining sins of omission and commission.

KULDEEP SINGH

1. ABSTRACT

The aim of this project is to design and implement a comprehensive methodology for Advanced Persistent Threat (APT) detection and automated incident response using Wazuh Endpoint Detection and Response (EDR) integrated with n8n workflow automation. Advanced Persistent Threats are stealthy, long-term cyberattacks that leverage sophisticated techniques to evade traditional security mechanisms, posing a significant risk to modern enterprise networks and critical infrastructures. This project focuses on identifying APT behaviors through continuous monitoring, log analysis, threat correlation, and real-time alerting.

The proposed methodology integrates multiple detection techniques, including host-based intrusion detection, log analysis, file integrity monitoring, vulnerability detection, and behavioral analysis provided by Wazuh EDR. Automation workflows built using n8n are employed to orchestrate incident response actions such as alert enrichment, threat classification, notification, containment, and remediation. By simulating real-world APT attack scenarios, the system evaluates the effectiveness of detection rules and automated response mechanisms.

The study analyzes the security architecture of enterprise environments, including endpoints, servers, network devices, and applications, along with the communication protocols and logging mechanisms involved. Emphasis is placed on detecting lateral movement, privilege escalation, persistence mechanisms, and command-and-control activities commonly associated with APTs. The project also addresses challenges related to scalability, real-time response, alert fatigue, and interoperability between security tools.

The outcome of the project is a structured and scalable framework for APT detection and automated incident response, enhancing organizational security posture by reducing detection time and response latency. This work contributes to the development of intelligent, resilient, and automated security operations capable of countering modern cyber threats in dynamic environments.

Keywords – *Advanced Persistent Threat (APT), Endpoint Detection and Response (EDR), Wazuh, Automated Incident Response, n8n, Cybersecurity*

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4. Abbreviations & Acronyms

Table 1: Acronyms & Full forms

Sr. No.	Acronym	Full Form
1	APT	Advanced Persistent Threat
2	EDR	Endpoint Detection and Response
3	SIEM	Security Information and Event Management
4	SOAR	Security Orchestration, Automation, and Response
5	Wazuh	Open-source Security Monitoring and EDR Platform
6	n8n	Node-based Workflow Automation Tool
7	MITRE ATT&CK	Adversarial Tactics, Techniques, and Common Knowledge
8	C2	Command and Control
9	FIM	File Integrity Monitoring
10	IOC	Indicator of Compromise
11	UEBA	User and Entity Behavior Analytics

12	CVE	Common Vulnerabilities and Exposures

5. Introduction

5.1 Background

The rapid evolution of the Internet and digital technologies has significantly transformed the way organizations operate, communicate, and store information. Modern enterprises increasingly rely on interconnected systems, cloud services, and distributed endpoints, making cyberspace an essential component of daily operations. While this digital transformation improves efficiency and scalability, it also expands the attack surface, enabling cyber adversaries to launch sophisticated and long-lasting attacks**.**

Among modern cyber threats, Advanced Persistent Threats (APTs) represent one of the most critical and dangerous categories. APTs are designed to remain undetected within a target environment for extended periods while silently stealing sensitive information or disrupting operations. These attacks pose serious risks to enterprises, governments, and critical infrastructure.

5.2 Advanced Persistent Threats (APTs)

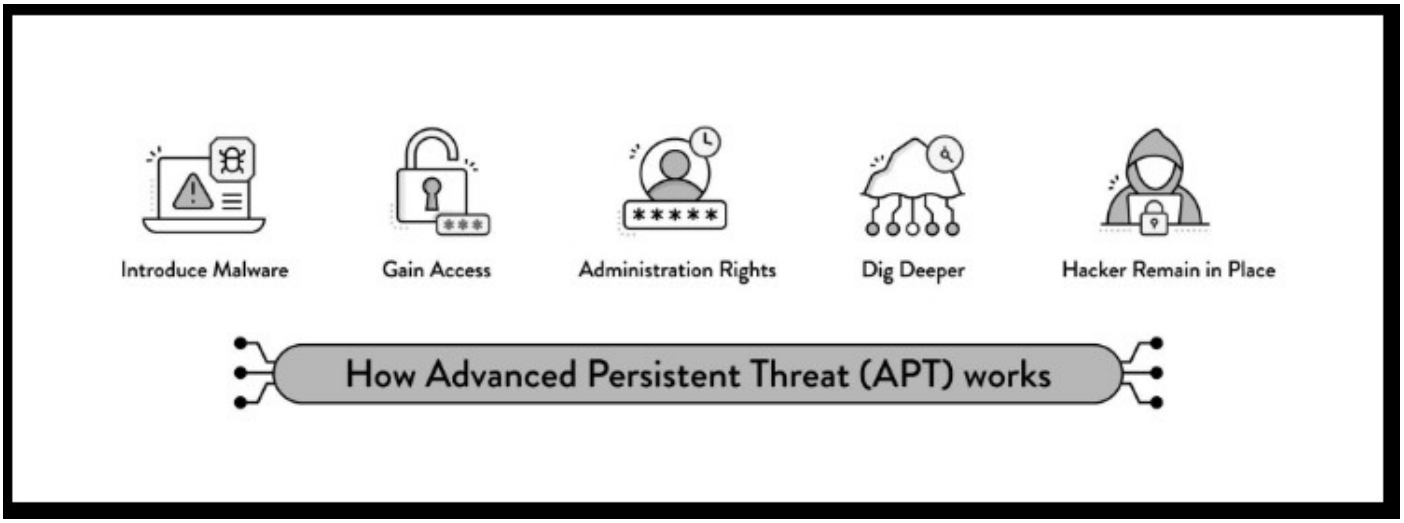


Fig 1: APT working

Advanced Persistent Threats are targeted, stealthy, and continuous cyberattacks in which attackers gain

unauthorized access to a network and maintain long-term presence. Unlike conventional cyberattacks that focus on immediate damage, APTs pursue long-term objectives such as data exfiltration, espionage, surveillance, and system manipulation.

APTs leverage a combination of advanced techniques including social engineering, malware, privilege escalation, lateral movement, and command-and-control (C2) communication. These techniques allow attackers to blend malicious activity with legitimate system behavior, making detection extremely challenging for traditional security solutions.

5.3 Increasing Risk of APT Attacks

With the widespread adoption of enterprise networks, cloud infrastructures, remote work environments, and Internet-connected endpoints, the risk of APT attacks has increased significantly. Attackers exploit software vulnerabilities, system misconfigurations, weak authentication mechanisms, and human errors to infiltrate networks.

Many organizations lack real-time visibility into endpoint activities and fail to correlate security events across systems effectively. As a result, APT attacks often remain undetected for months, leading to severe financial losses, operational disruption, and reputational damage.

5.4 Role of Endpoint Detection and Response (EDR)

Endpoint Detection and Response (EDR) has emerged as a critical security mechanism for defending against APTs. EDR solutions continuously monitor endpoint behavior, collect detailed logs, and analyze activities to detect suspicious patterns.

Wazuh, an open-source EDR and Security Information and Event Management (SIEM) platform, provides capabilities such as log analysis, file integrity monitoring, vulnerability detection, threat intelligence integration, and behavioral analysis. While Wazuh enables effective detection, detection alone is insufficient without timely and coordinated incident response.

5.5 Need for Automated Incident Response

To minimize the impact of APT attacks, rapid and coordinated incident response is essential. Manual incident handling is time-consuming, error-prone, and contributes to alert fatigue in Security Operations Centers (SOCs).

n8n, an open-source workflow automation platform, enables the automation of incident response actions such as alert enrichment, threat classification, notification, logging, and controlled containment. Integrating Wazuh with n8n allows security events to be processed intelligently, significantly reducing response time and manual effort.

5.6 Purpose of the Project

The purpose of this project is to design and implement an automated system for Advanced Persistent Threat detection and incident response using Wazuh EDR and n8n. The system aims to detect stealthy, long-term cyberattacks by continuously monitoring endpoint and network activities, correlating security events, and identifying suspicious behavior patterns.

By integrating Wazuh's detection capabilities with n8n's automation features, the project enhances SOC efficiency by automating alerting, incident logging, and optional containment actions, ensuring faster and more reliable handling of APT-related security incidents.

5.7 Scope of the Project

The scope of this project includes endpoint monitoring, centralized log collection, threat detection, alert correlation, and automated incident response within an enterprise-like environment. It involves deploying Wazuh agents on endpoints, performing centralized analysis through the Wazuh Manager, and

automating detection and response workflows using n8n.

The system focuses on detecting activities such as authentication failures, suspicious process execution, persistence mechanisms, privilege escalation, lateral movement, and abnormal network connections. Automated containment actions are kept optional to reduce false positives. The implementation is suitable for educational, laboratory, and small-to-medium enterprise SOC environments.

5.8 Definition and Core Characteristics of APT

An Advanced Persistent Threat (APT) is a highly sophisticated and targeted cyberattack in which adversaries gain unauthorized access to a network and remain undetected for long periods—often months or years. These attacks are meticulously planned and executed by skilled and well-funded actors such as nation-states, organized cybercriminal groups, or hacktivists.

Key characteristics of APTs include advanced techniques, long-term persistence, targeted and stealthy operations, and systematic data exfiltration. Attackers frequently use living-off-the-land techniques, encrypted communication channels, and covert data transfer mechanisms to evade detection.

5.9 APT Lifecycle

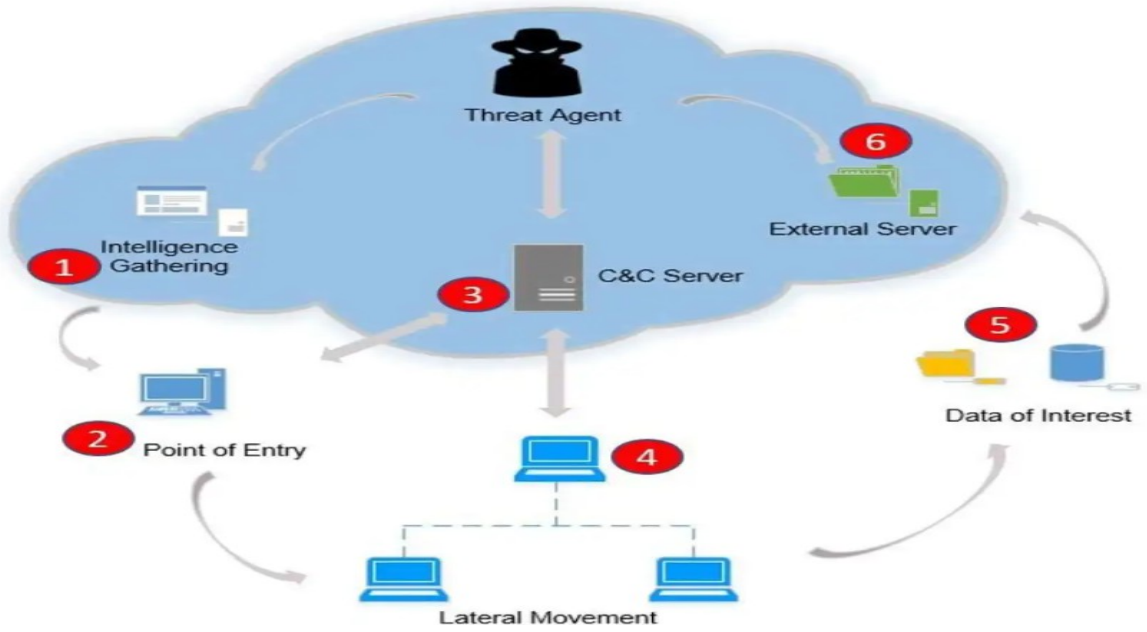


Fig 2: APT Lifecycle

Advanced Persistent Threats follow a structured, multi-stage lifecycle:

Reconnaissance:

Attackers gather detailed information about the target using OSINT, scanning, and social engineering.

Initial Compromise:

Entry is gained through phishing, malware, stolen credentials, or vulnerability exploitation.

Establishing Persistence:

Backdoors, scheduled tasks, registry changes, or malicious services ensure long-term access.

Privilege Escalation:

Attackers obtain elevated privileges to access critical resources.

Lateral Movement:

The attack spreads across systems to identify high-value targets.

Command and Control (C2):

Compromised systems communicate with attacker-controlled servers.

Data Exfiltration:

Sensitive data is compressed, encrypted, and exfiltrated gradually.

Maintaining Presence:
Logs are altered and activities are disguised to evade detection.

The proposed system using Wazuh and n8n addresses each stage through continuous monitoring, correlation, and automated response.

5.10 Significance of the Project

This project highlights the importance of proactive monitoring, automation, and intelligent incident response in combating Advanced Persistent Threats. By integrating EDR with workflow automation, the proposed framework improves detection accuracy, reduces response time, and strengthens the overall security posture of modern enterprise environments.

6. OPERATING SYSTEM REQUIREMENTS

Table 2: Operating system & Purpose

Sr. No.	Operating System	Purpose
1	Linux (Ubuntu / Kali Linux / CentOS)	Used for deploying Wazuh Manager, Wazuh Indexer (OpenSearch), Wazuh Dashboard, and n8n automation workflows for centralized monitoring and automated incident response.
2	Windows (Windows 10 / Windows Server)	Used as endpoint systems where Wazuh agents are installed to monitor user activity, authentication events, registry changes, and suspicious APT-related behavior.

7. SOFTWARE REQUIREMENTS

Table 3: Software & Role in project

Sr. No.	Software	Role in Project
1	Wazuh EDR	Endpoint detection, log analysis, event correlation, file integrity monitoring, vulnerability detection, and APT detection.
2	n8n	Automation of incident response workflows including alert processing, filtering, notification, and orchestration.
3	OpenSearch (Wazuh Indexer)	Indexing, storing, and retrieving large volumes of security logs and alerts.
4	Web Browser (Chrome / Firefox)	Accessing Wazuh Dashboard and n8n web interface for monitoring and management.
5	Email & Messaging Services (SMTP / Slack API)	Used for real-time alert notifications and incident escalation.

8. Literature Survey

Advanced Persistent Threats (APTs) represent a class of sophisticated cyberattacks that are designed to remain undetected within a target environment for extended periods. A detailed survey conducted by Ghafir and Prenosil [1] provides a comprehensive understanding of APTs by analyzing their complete attack lifecycle. The study explains how APT attacks progress through multiple stages, including initial compromise, privilege escalation, lateral movement, persistence, and data exfiltration. The authors highlight that traditional security mechanisms such as signature-based intrusion detection systems are inadequate for detecting APTs due to their stealthy behavior and ability to evade known signatures. The paper emphasizes the importance of behavior-based detection, continuous monitoring, and correlation of security events across multiple layers, thereby motivating the use of advanced detection frameworks such as EDR and SIEM systems.

Endpoint Detection and Response (EDR) has emerged as a key technology for detecting advanced cyber threats at the endpoint level. Chuvakin, Schmidt, and Phillips [2] present a systematic review of EDR technologies, focusing on their capabilities and operational challenges. The study explains how EDR solutions continuously monitor endpoint activities, collect detailed logs, and analyze behavioral patterns to detect suspicious activities associated with advanced attacks. It also discusses major challenges such as alert fatigue, scalability issues, and delayed incident response caused by manual analysis. The authors conclude that EDR solutions must be integrated with automation and orchestration mechanisms to improve response efficiency. This study supports the adoption of Wazuh EDR as a core detection component and highlights the need for automated incident response using tools such as n8n.

Log analysis and event correlation play a crucial role in identifying APT behavior that may not be evident through individual alerts. Zeng and Hu [3] investigate the detection of APTs using log analysis and machine learning techniques. Their research demonstrates that anomalies related to lateral movement, privilege escalation, and persistence can be detected by correlating multiple low-level events over time. The study emphasizes that centralized log collection and correlation are essential for identifying complex and stealthy attack patterns. These findings reinforce the relevance of Wazuh EDR's centralized log analysis and correlation capabilities in detecting Advanced Persistent Threats.

Based on the reviewed literature, it is evident that while EDR and log-based detection mechanisms are effective in identifying APT activities, many existing approaches lack automated incident response. The

integration of Wazuh EDR with n8n automation in the proposed project addresses this gap by enabling real-time detection, correlation, and automated response to Advanced Persistent Threats.

9. Motivation

Table 4: Motivation factor & Description

Sr. No.	Motivation Factor	Description
1	Increase in Advanced Persistent Threats	APT attacks are stealthy, targeted, and long-term, making them difficult to detect using traditional security solutions.
2	Limitations of Conventional Security Systems	Traditional security systems often fail to detect multi-stage and low-and-slow attacks effectively.
3	Impact on Organizations	APT attacks can lead to data breaches, financial losses, reputational damage, and operational disruption.
4	Need for Automation	Manual incident response is slow and error-prone; automation improves response time and accuracy.
5	Industry Demand and Practical Relevance	Growing demand for professionals skilled in EDR, SIEM, and automated incident response.

10. Problem Statement

Advanced Persistent Threats (APTs) are stealthy and long-term cyberattacks that are difficult to detect using traditional security solutions. Modern enterprise environments generate large volumes of security data, making manual monitoring and incident response inefficient and prone to delays. Existing systems often lack effective integration between threat detection and response mechanisms. Therefore, there is a need for an automated framework that can efficiently detect APT activities and respond to incidents in real time. This project aims to address this challenge by using Wazuh EDR for threat detection and n8n for automated incident response to enhance security and reduce response time

11. Wazuh as an Endpoint Detection and Response (EDR) Tool

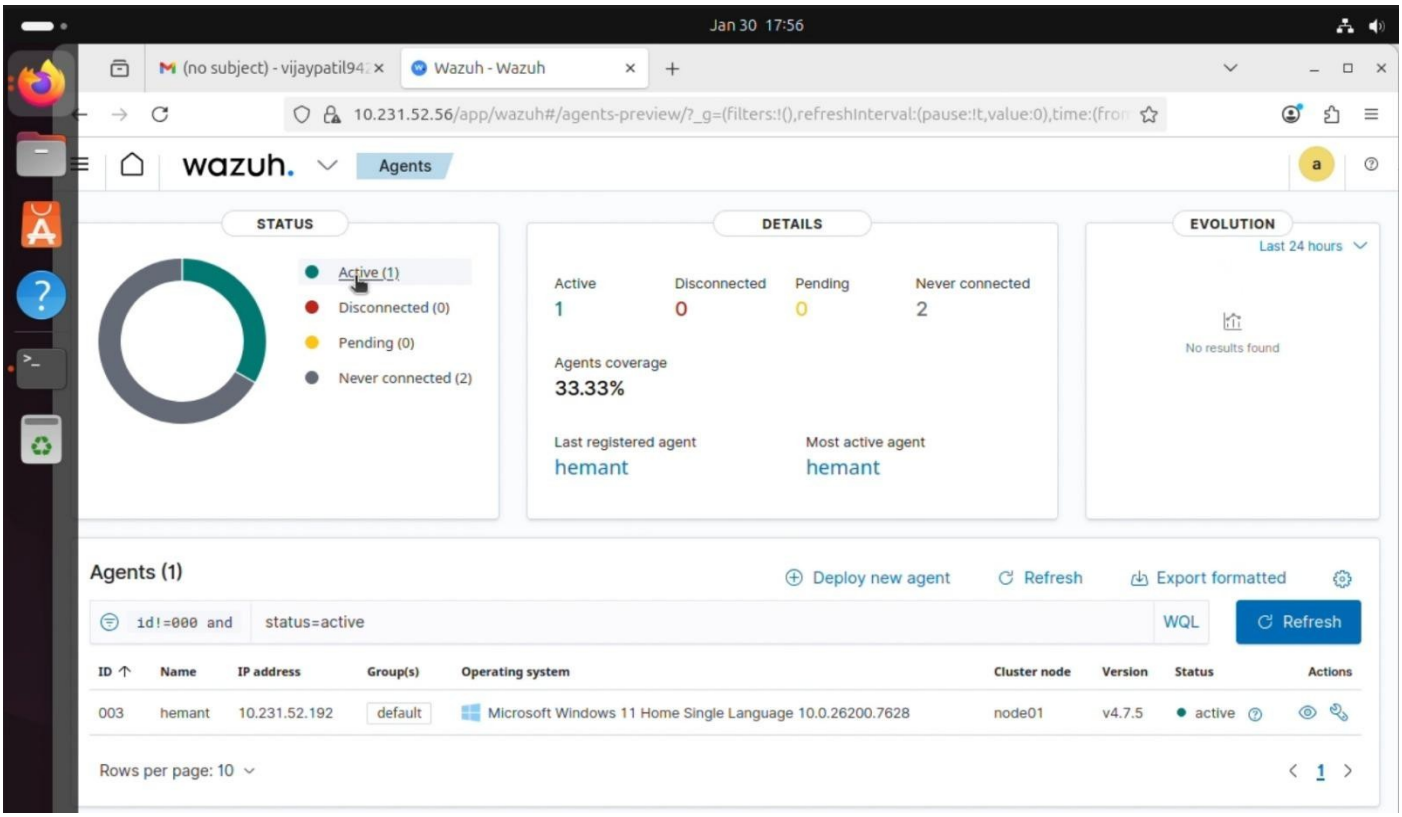


Fig 3: Wazuh Agents staus

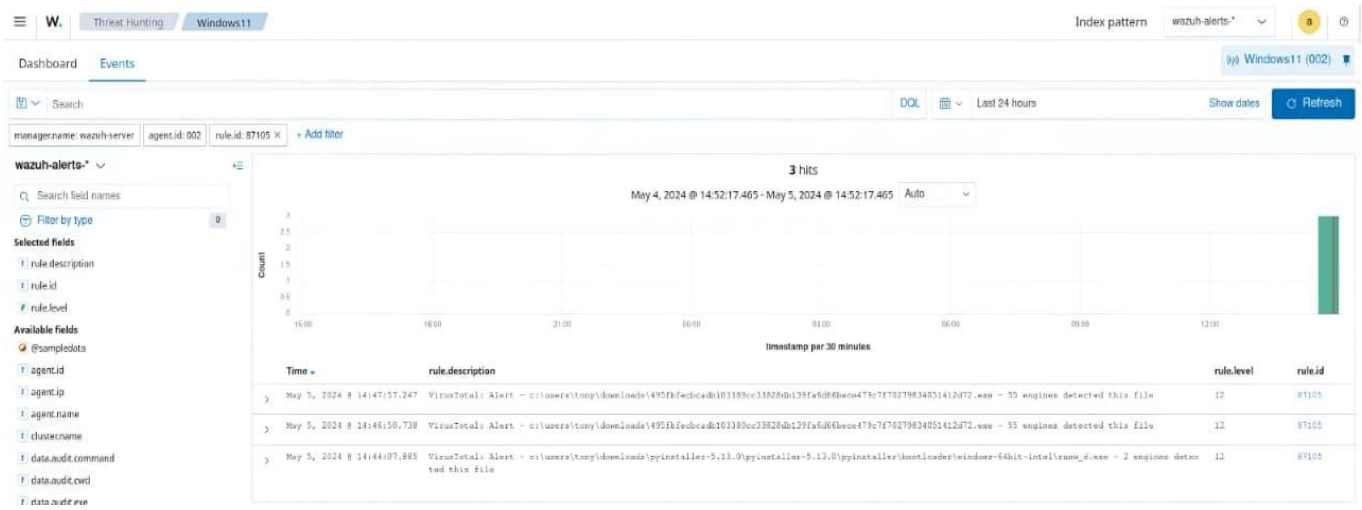


Fig 4: Wazuh logs

11.1 Overview of Wazuh EDR in the Proposed System

Wazuh is used in this project as the primary Endpoint Detection and Response (EDR) platform. It provides centralized visibility into endpoint activity, continuous log collection, security event correlation, and alert generation. In the context of Advanced Persistent Threat (APT) detection, Wazuh plays a crucial role by monitoring endpoints over long periods and identifying stealthy attack behavior that may otherwise remain unnoticed.

The first screenshot shows the Wazuh Modules dashboard, where the overall status of deployed agents is visible. The dashboard displays the total number of agents, active agents, disconnected agents, pending agents, and never-connected agents. This view is important because APT detection depends heavily on continuous endpoint visibility. If an endpoint is disconnected or never connected, it becomes a blind spot for attackers to exploit.

In the shown environment, three agents are registered, with one active agent and others in a never-connected state. This confirms that the Wazuh Manager is operational and capable of receiving data from endpoints.

11.2 Agent Deployment and Endpoint Registration

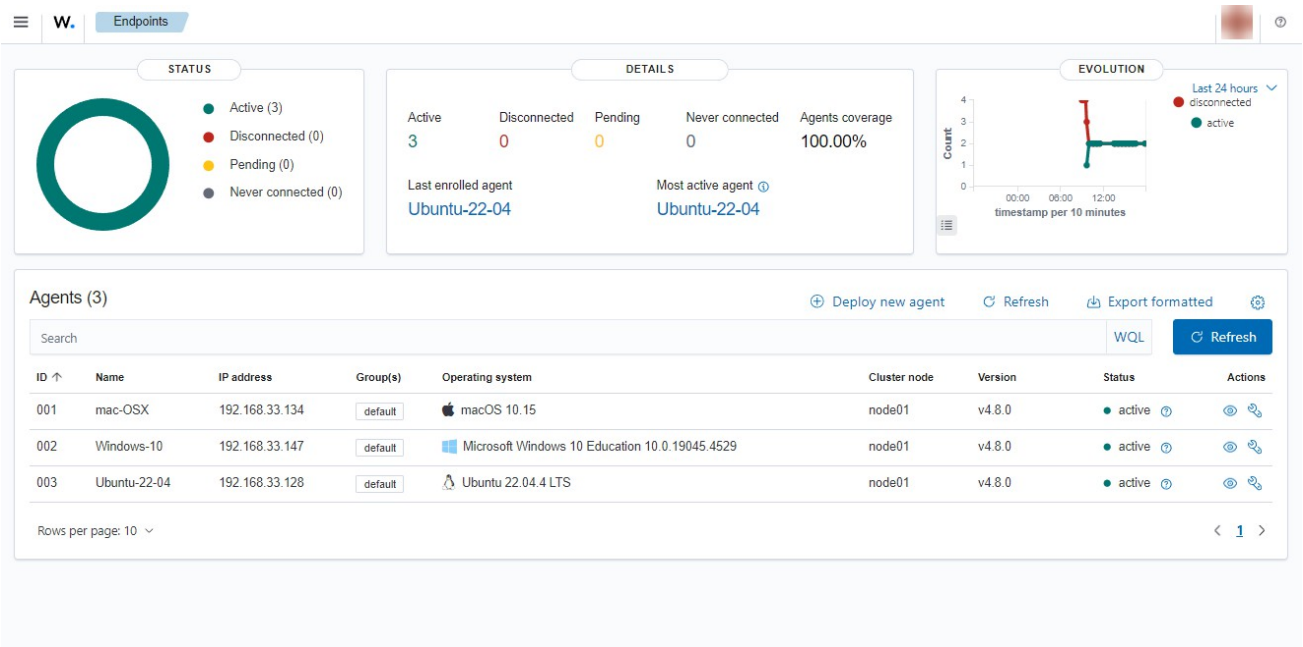


Fig 5 Wazuh Enpoints (connected IP Address)

Once Wazuh is installed, agents are deployed on monitored systems such as Windows 11 and Linux (Kali/Ubuntu) machines. The second screenshot displays the Agents section, where detailed information about each registered endpoint is shown. This includes agent ID, hostname, IP address, operating system, Wazuh version, and agent status.

The screenshot clearly shows a Windows 11 endpoint registered under the name *hemant*, marked as active. This confirms successful agent installation, authentication with the Wazuh Manager, and secure communication between the agent and server. In APT detection, this step is critical because all subsequent monitoring, detection, and response actions rely on accurate endpoint telemetry.

The agent continuously sends logs related to authentication attempts, system processes, file changes, and network activity to the Wazuh Manager.

11.3 Security Event Collection and Malware Detection

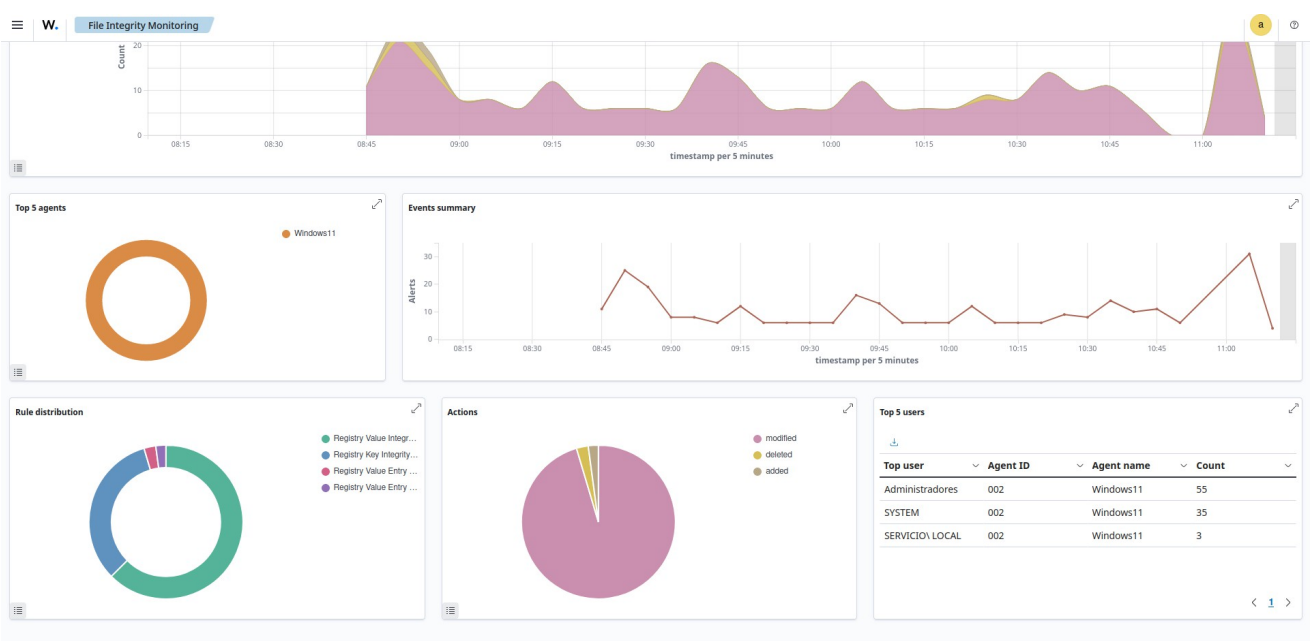


Fig 6: Wazuh file integrity monitering

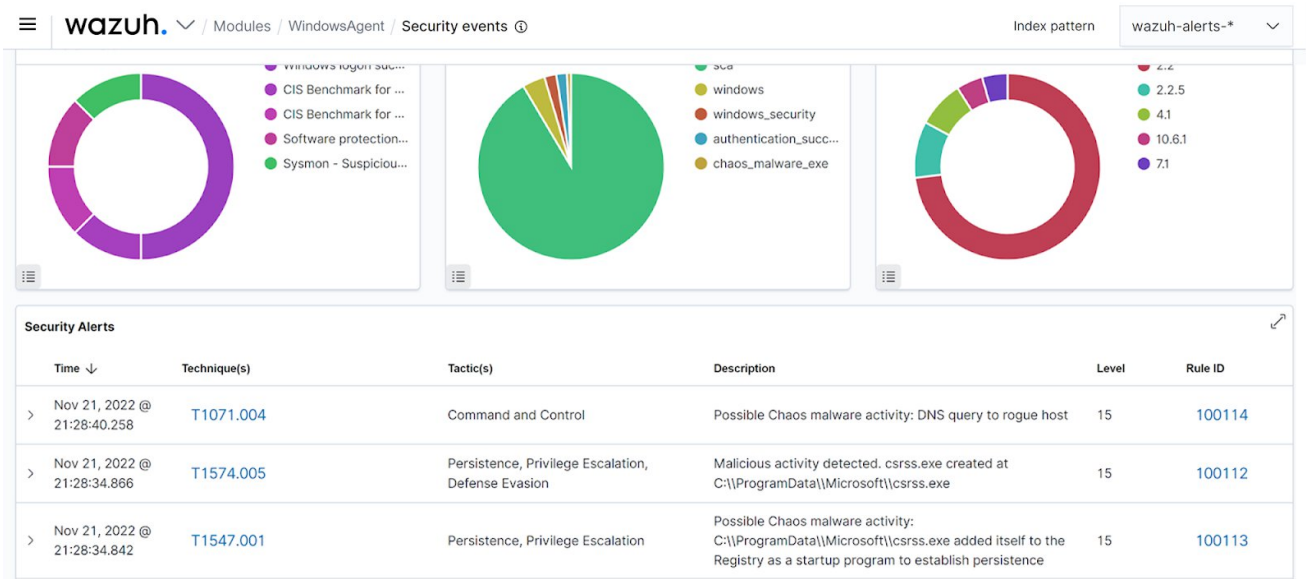


Fig 7: Security events

After agents are active, Wazuh begins collecting and analyzing security events. The third screenshot shows security events generated for a Windows endpoint, including alerts where malicious files were detected by multiple antivirus engines. These alerts are generated using Wazuh's rule engine, which correlates endpoint activity with known malware signatures and behavioral indicators.

From an APT detection perspective, such alerts are important early indicators of initial compromise. Even if malware is not immediately destructive, it may be part of a larger APT campaign attempting to establish a foothold in the system.

Each alert includes detailed metadata such as rule ID, severity level, timestamp, agent name, and description, which later helps in correlating multi-stage attacks.

11.4 Threat Hunting and Behavioral Analysis

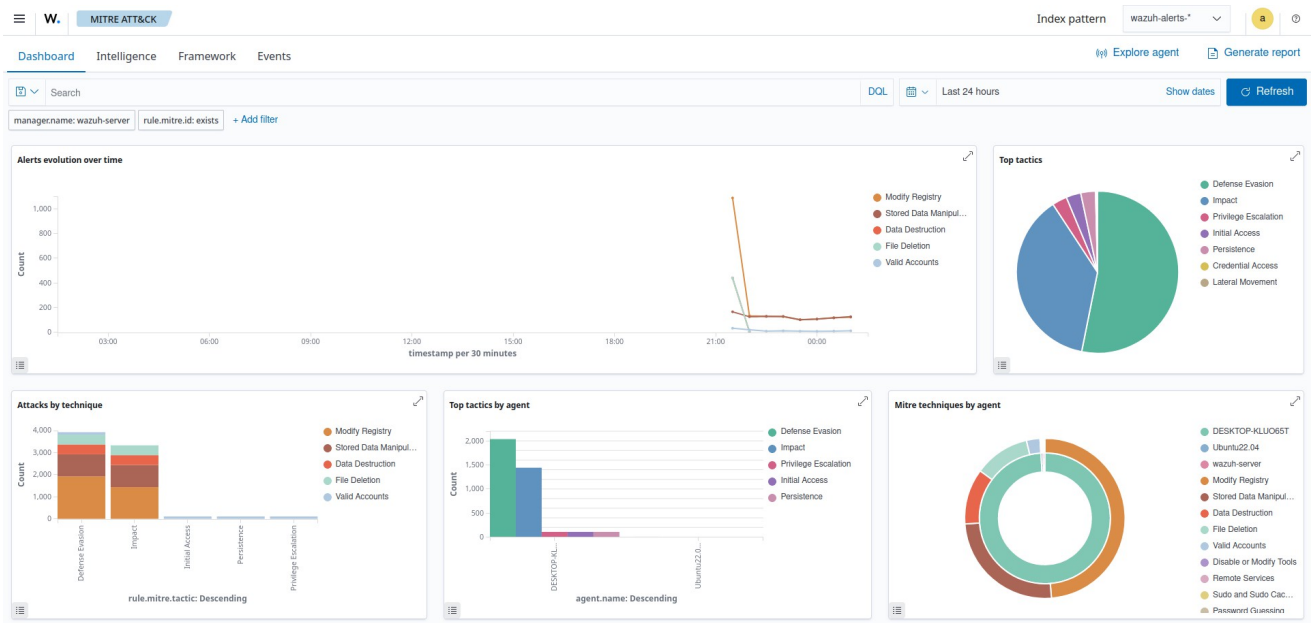


Fig 8: Mitre Attack

The next screenshot represents the Threat Hunting dashboard, which aggregates and visualizes large volumes of security data. It displays total alerts, high-severity alerts, authentication failures, and successful logins over a given time period.

This view is especially valuable for detecting APT patterns, as APTs rarely generate a single high-severity alert. Instead, they produce repeated low-level anomalies such as failed login attempts, suspicious access times, or abnormal authentication behavior. By observing spikes and trends in these graphs, analysts can identify suspicious activity spanning multiple hours or days.

The presence of hundreds of authentication failures in the dashboard indicates potential brute-force attempts or credential misuse — common techniques used during the early and lateral movement phases of APT attacks.

11.5 Centralized Log Correlation and Visualization

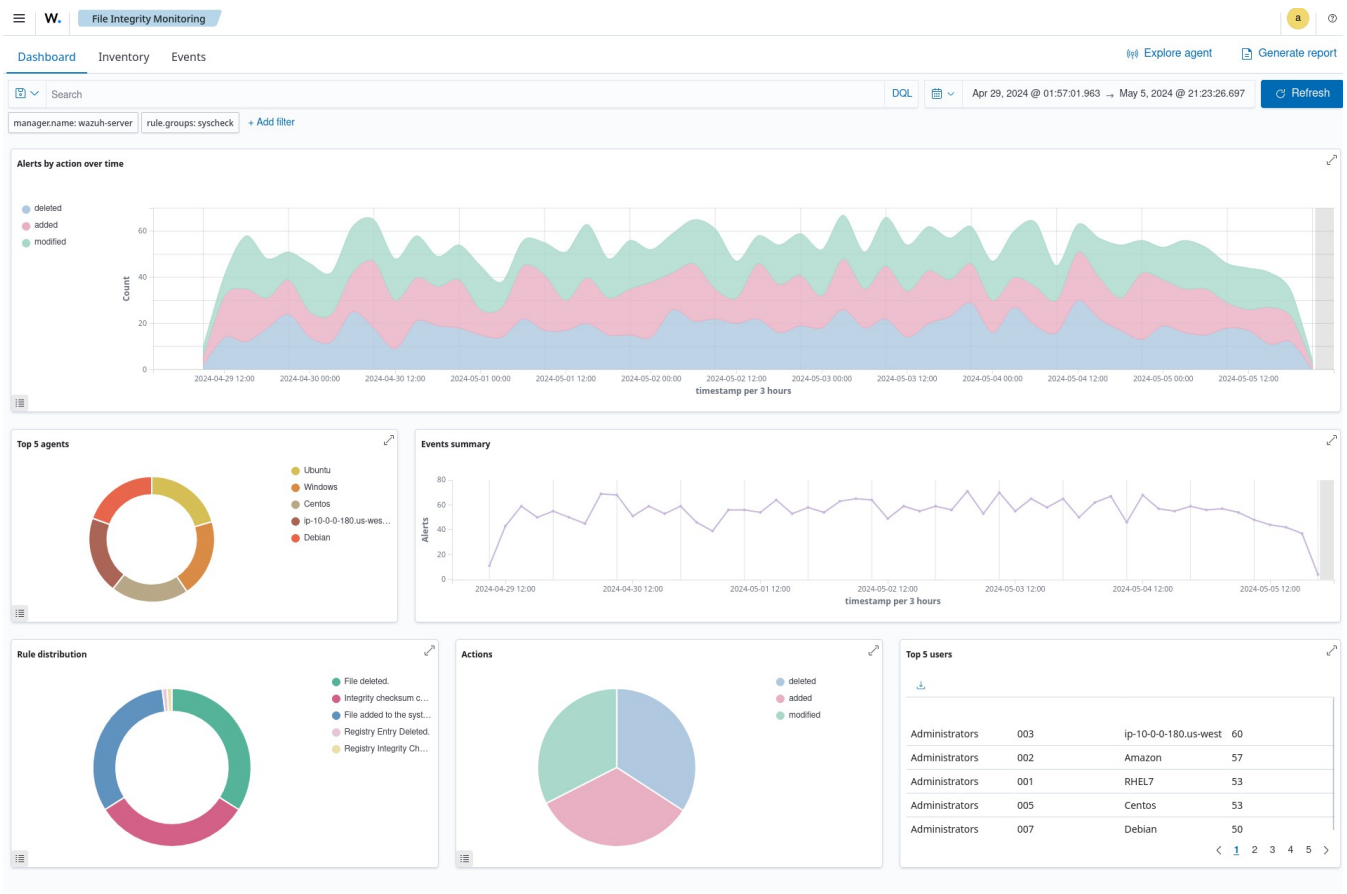


Fig 9: Wazuh monitoring graph

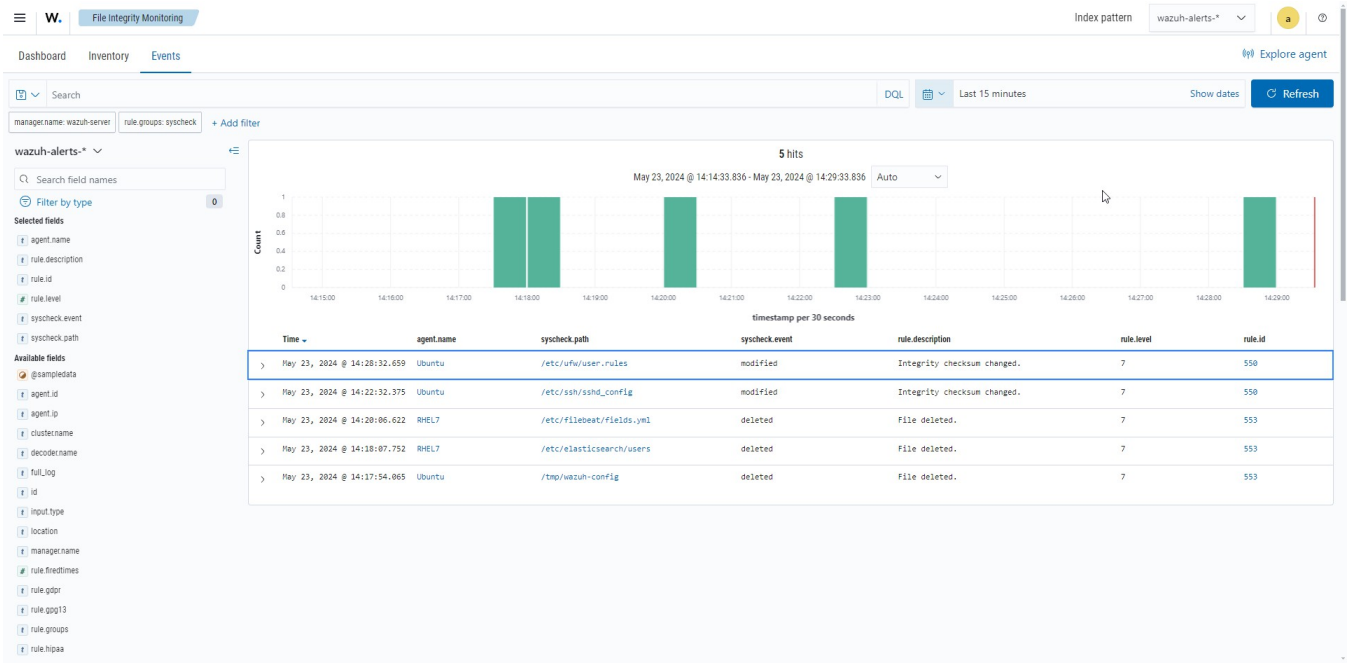


Fig 10: File monitoring detail

Another screenshot shows the Security Information Management and Auditing modules within Wazuh. These modules provide access to integrity monitoring, policy compliance, and system auditing data. File Integrity Monitoring (FIM) is especially important for detecting persistence mechanisms, such as unauthorized changes to startup scripts, registry keys, or system binaries.

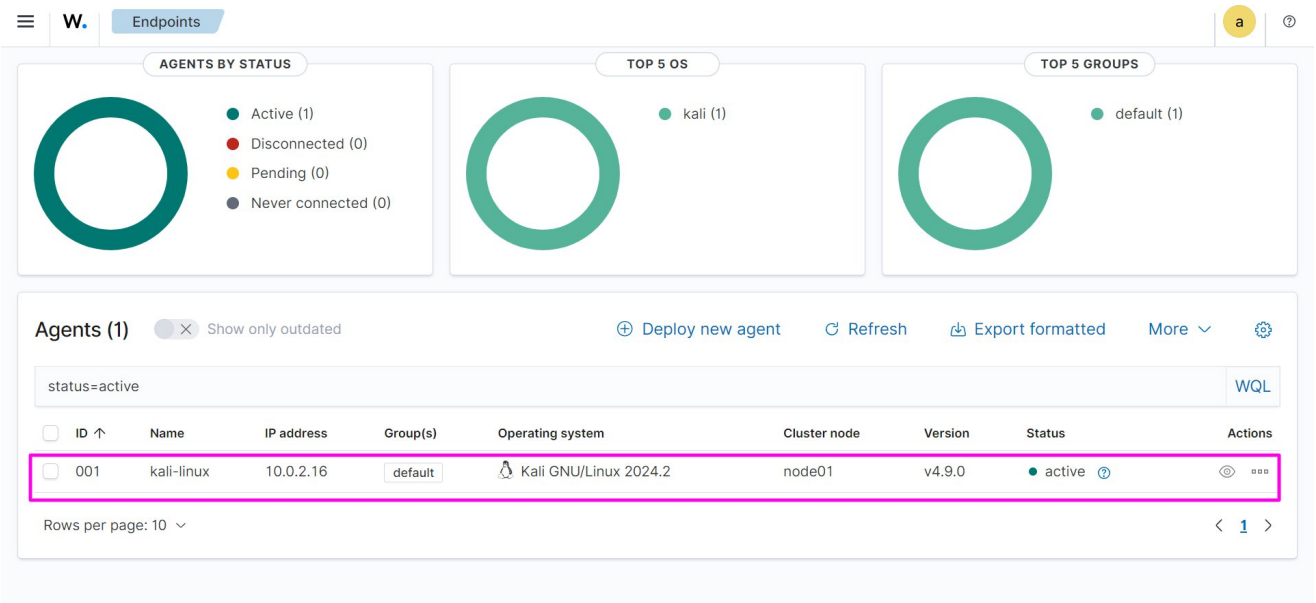
For APT detection, these capabilities ensure that even if attackers attempt to hide their tracks by modifying system files, such actions are recorded, correlated, and preserved for analysis.

11.6 Wazuh API Configuration and Integration Readiness

The screenshot showing the Wazuh API configuration confirms that the REST API is enabled and online. This API plays a critical role in the project because it allows external systems, such as n8n, to fetch alerts programmatically.

Through this API, high-severity alerts related to APT activities are retrieved, enriched, and forwarded to automated response workflows. Without this integration layer, incident response would remain manual and delayed.

11.7 Multi-Platform Monitoring (Windows and Linux)



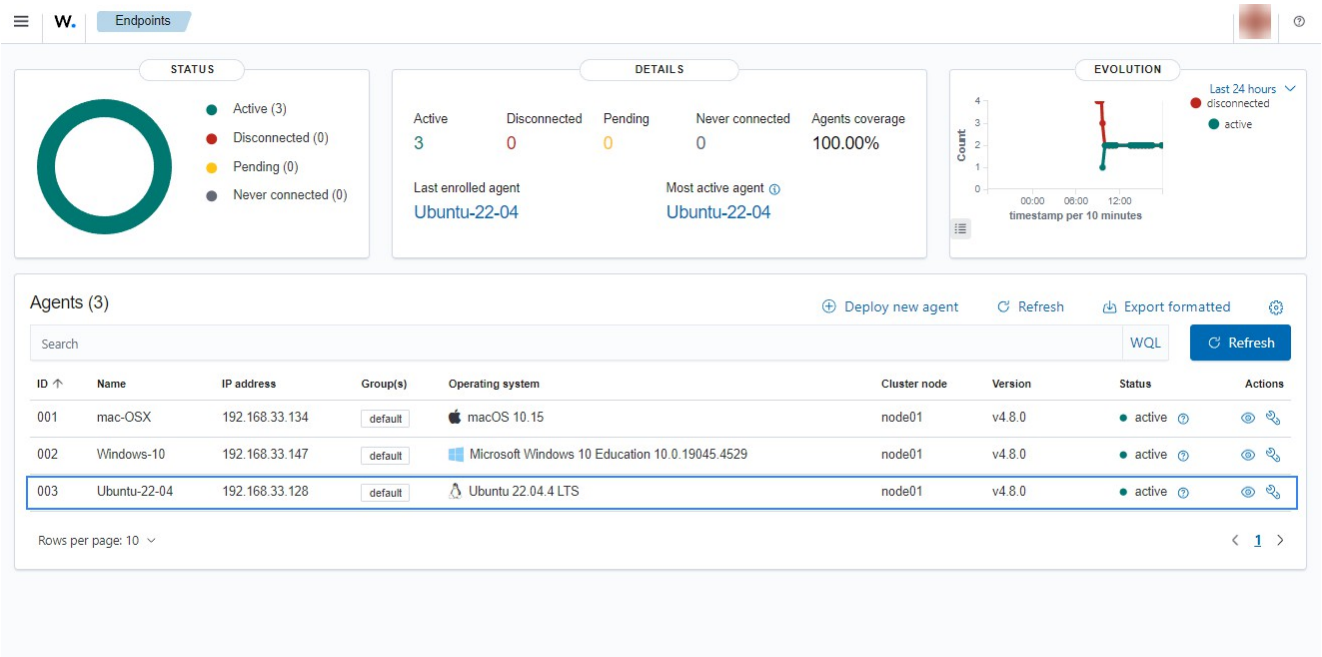


Fig 11: Endpoints information

The final screenshots show Linux-based endpoints (Kali Linux) successfully registered and monitored by Wazuh. This confirms that the system supports heterogeneous environments, which is a key requirement for enterprise-level APT detection.

Attackers often move laterally between different operating systems. Wazuh's ability to monitor both Windows and Linux endpoints ensures visibility across the entire infrastructure, enabling detection of

lateral movement and coordinated attacks.

11.8 Overall Process Flow in the Context of APT Detection

In summary, the Wazuh EDR process in this project follows a continuous and structured flow. Endpoints are first registered using agents, after which logs and system events are continuously collected. These events are analyzed using decoders and correlation rules mapped to MITRE ATT&CK techniques. Suspicious activities generate alerts that are visualized in dashboards and made available via the Wazuh API. This data is then consumed by n8n to perform automated incident response.

By combining long-term log retention, behavioral analysis, and centralized correlation, Wazuh provides a strong foundation for detecting Advanced Persistent Threats that rely on stealth, persistence, and gradual progression.

12. n8n as an Automation Platform and Its Integration with Wazuh

12.1 Overview of n8n in the Proposed System

n8n is an open-source workflow automation platform used in this project as the incident response and orchestration layer. While Wazuh EDR is responsible for detecting security threats and generating alerts, n8n automates the processing, enrichment, filtering, alerting, and response actions associated with APTs.

n8n provides a visual workflow interface combined with JavaScript-based logic, making it suitable for implementing security automation pipelines. Its ability to integrate with REST APIs, messaging platforms, email systems, databases, and security tools allows it to function as a lightweight SOAR (Security Orchestration, Automation, and Response) engine. In this project, n8n bridges the gap between detection and response by converting Wazuh alerts into actionable security operations.

12.2 Integration of n8n with Wazuh

Wazuh exposes detected security events and alerts through its REST API in a structured JSON format. n8n connects securely to this API using an HTTP request node with authentication tokens. This integration enables n8n to periodically retrieve new alerts related to endpoint and network activity.

By consuming alerts directly from Wazuh, n8n operates on real detection data rather than simulated inputs. This ensures that the automated workflow reflects realistic APT scenarios such as credential dumping, suspicious process execution, authentication abuse, and abnormal network behavior.

12.3 Step-by-Step Working of the n8n Workflow for APT Detection and Automated Incident Response

The n8n workflow is designed as a continuous security automation pipeline that transforms raw security telemetry into prioritized alerts and automated response actions. Each stage of the workflow plays a specific role in detecting and responding to Advanced Persistent Threats.

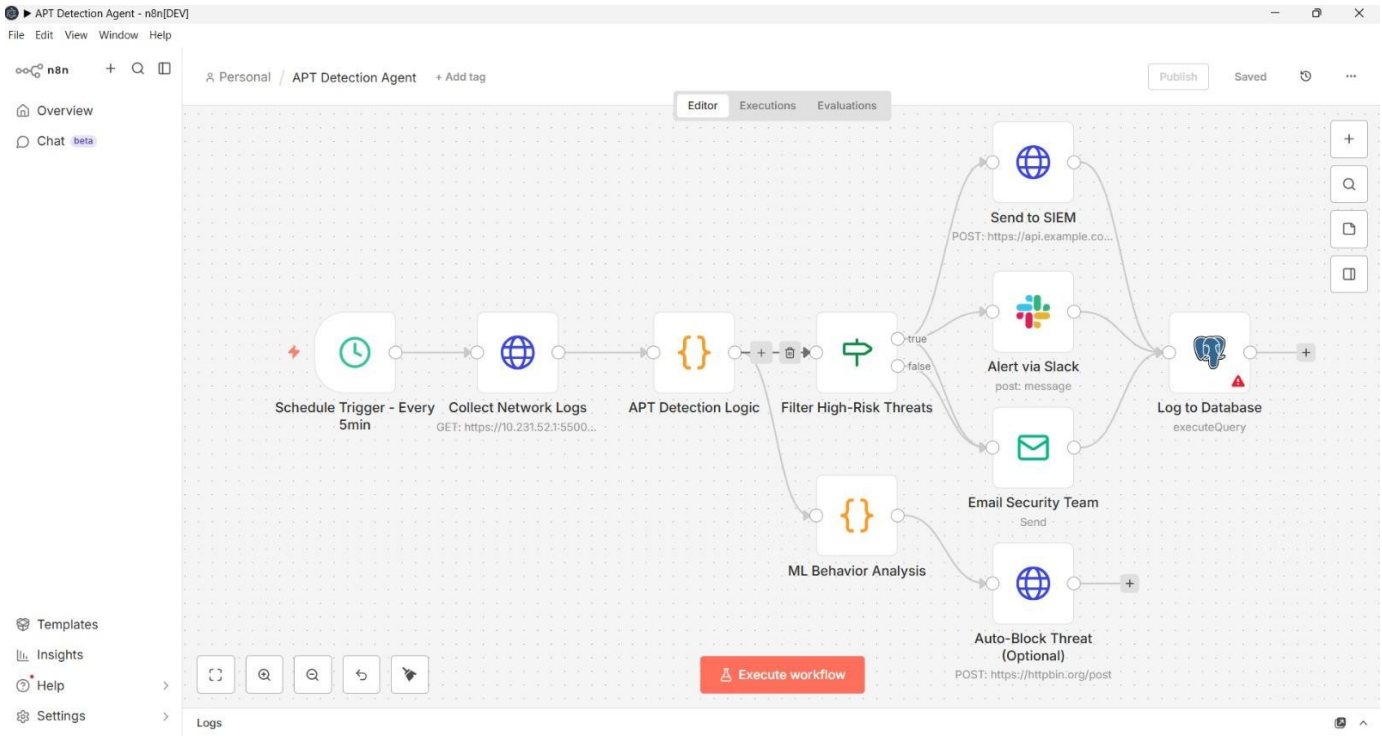


Fig 12: N8n workflow

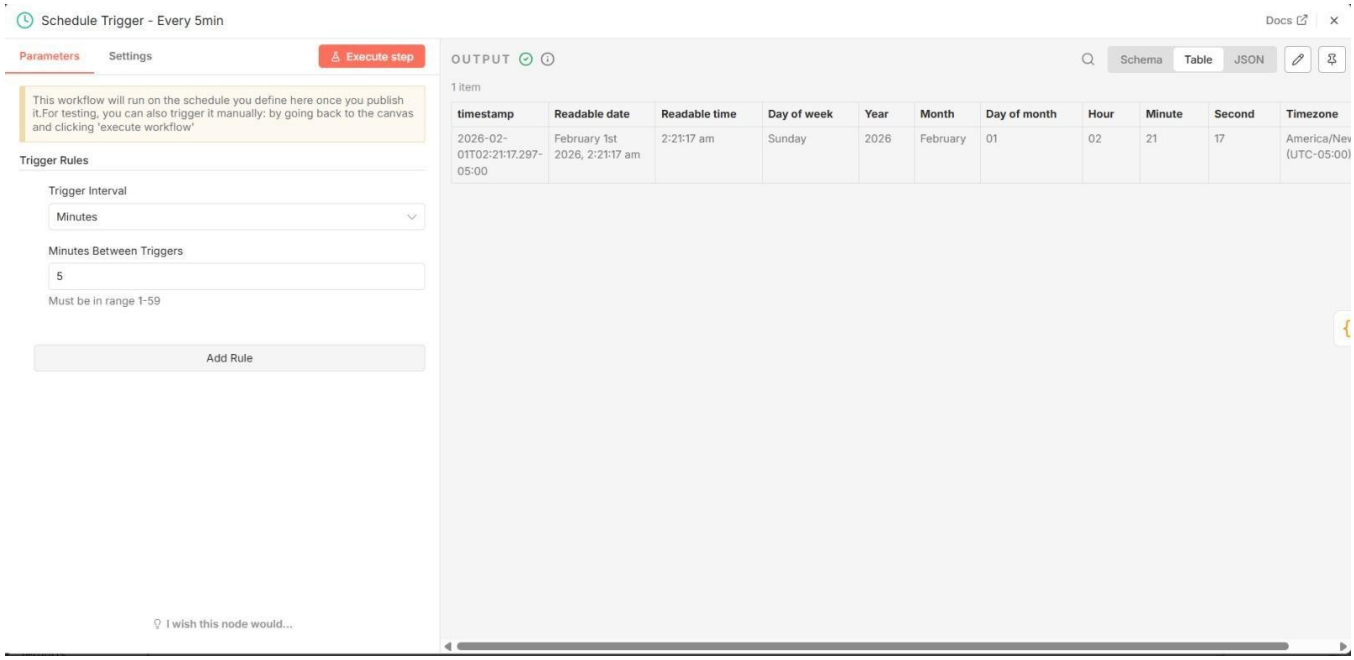


Fig 13: Schedule Trigger

This stage represents the Schedule Trigger, which acts as the starting point of the entire APT detection workflow. It automatically runs the n8n process every five minutes to ensure continuous monitoring without manual intervention. This guarantees that fresh security data is regularly pulled from Wazuh for analysis. As a result, the system maintains near real-time visibility of potential APT activities in the environment.

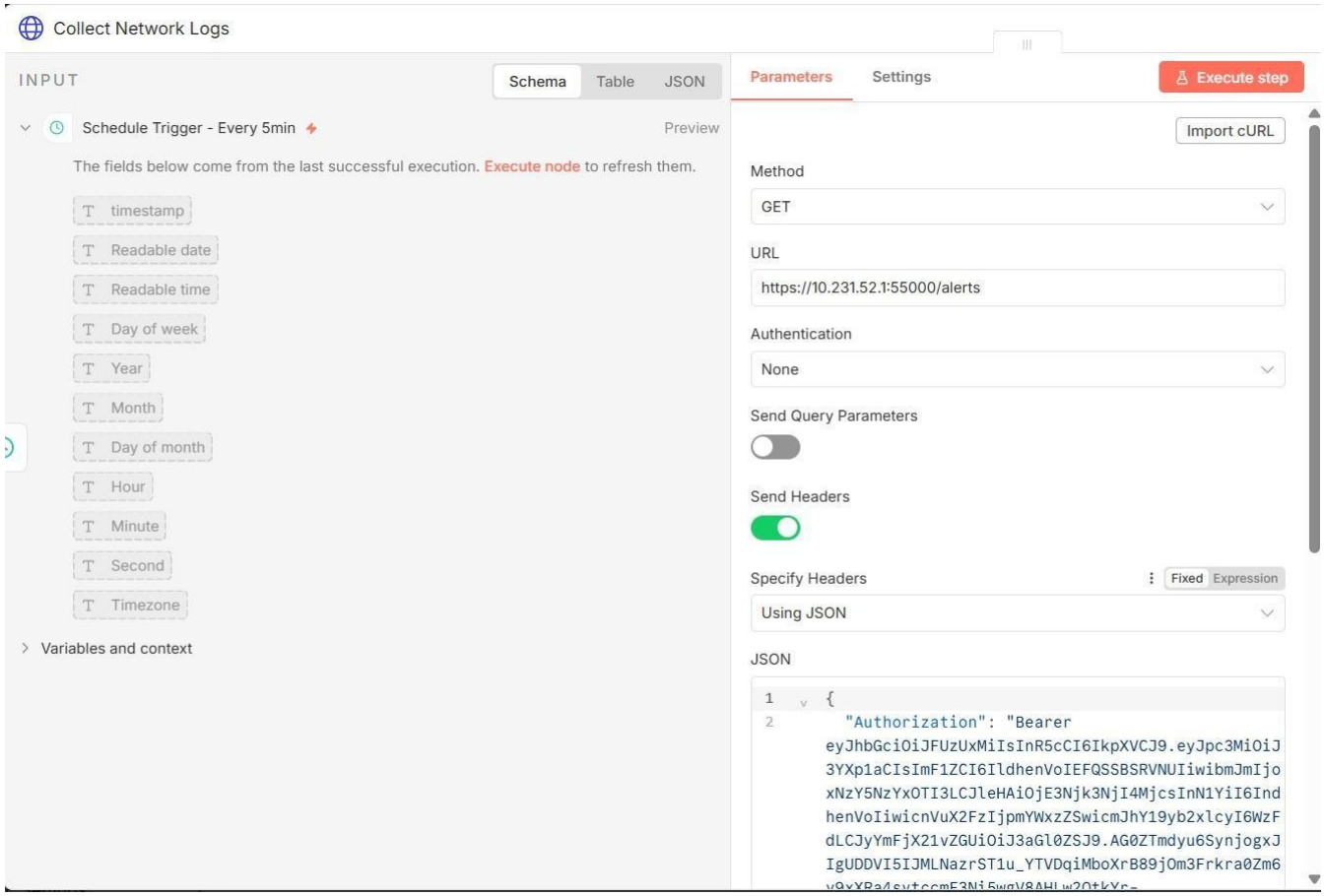


Fig 14: Collect Ntework logs

This stage represents the Collect Network Logs component, where n8n connects to the Wazuh manager through an HTTP API call to fetch real-time security alerts and event data. It periodically pulls logs related to network and endpoint activity, such as authentication attempts, suspicious connections, and system events. This step acts as the data ingestion layer of the workflow, ensuring that the APT detection process always works with fresh and updated security information. The collected logs then become the input for further correlation and threat analysis.

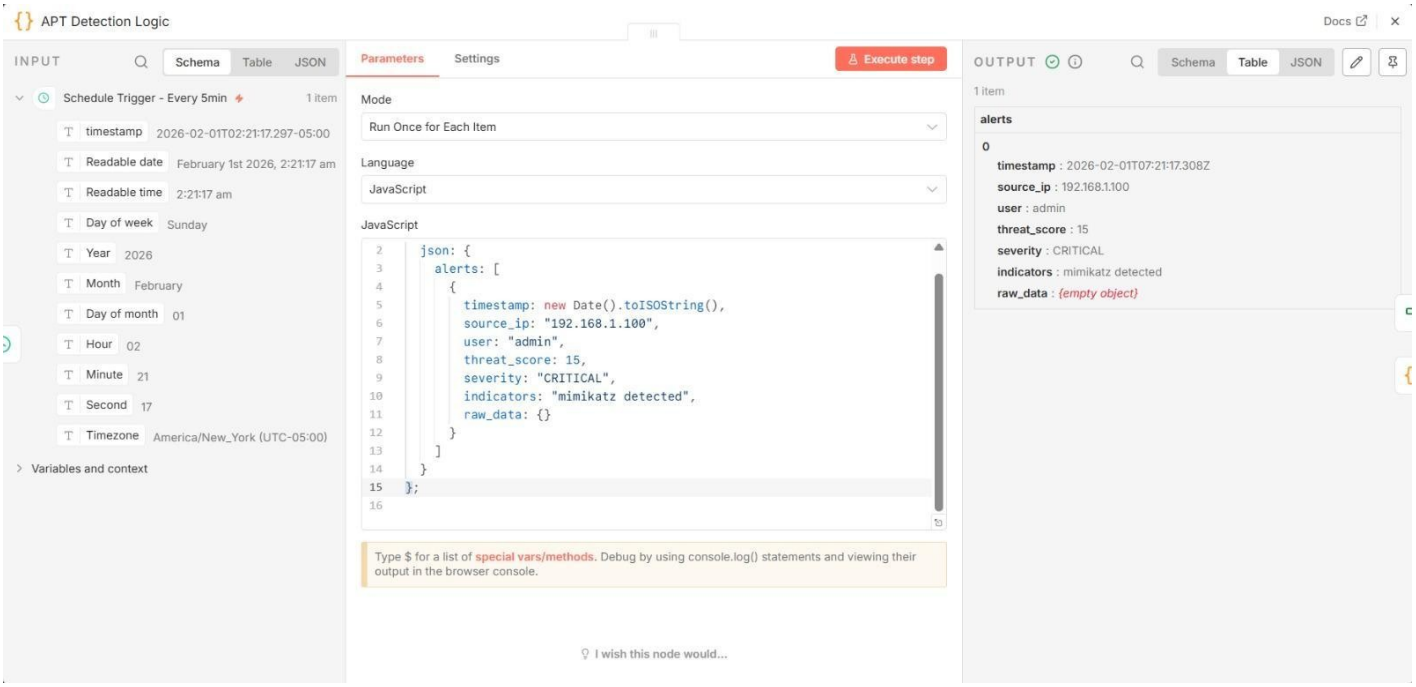


Fig 15: APT Dection Logic

This APT Detection Logic node acts as the core intelligence layer of the workflow, where incoming data is converted into meaningful security alerts. It uses JavaScript to define and simulate how Wazuh correlates events by assigning threat scores, severity levels, and attack indicators such as credential-dumping tools. This step represents the decision-making process of the SIEM, identifying whether observed activity matches known APT techniques. The output becomes a standardized alert format that drives all subsequent analysis, alerting, and response actions.

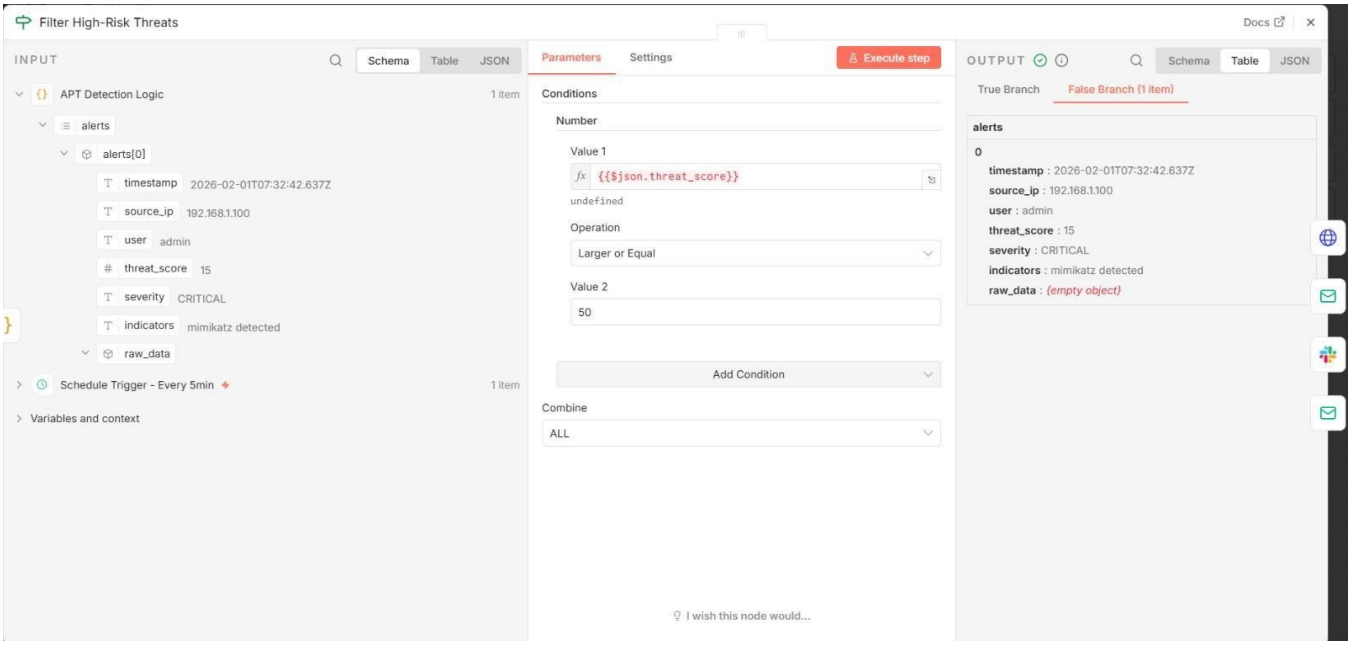


Fig 16: Filter High Risk Threats

This structure shows an automated APT detection workflow using Wazuh EDR/SIEM integrated with n8n for security automation. Wazuh collects and analyzes endpoint activity, generating structured alerts with details like threat score, severity, user, and indicators such as credential dumping behavior. These alerts are passed into n8n, where conditional logic evaluates the threat score against a defined risk threshold. High-risk alerts follow the true branch for automated response actions, while lower-risk events are filtered out to reduce noise. This setup enables efficient, real-time detection and response to advanced persistent threats.

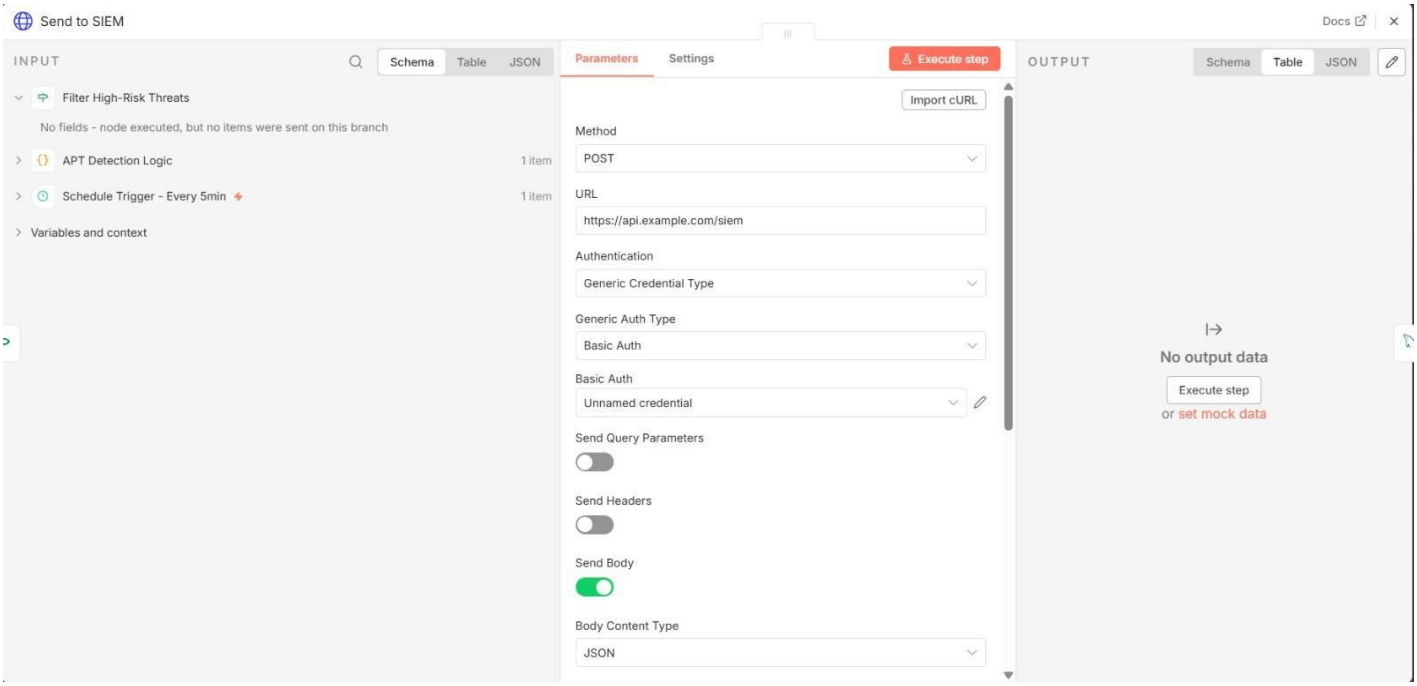


Fig 17: Send to SIEM

This step forwards high-risk APT alerts from n8n back into a central SIEM platform for correlation and long-term analysis. After Wazuh detects and n8n confirms a critical threat, the alert data is sent via a secure HTTP POST request in JSON format. Authentication ensures only trusted systems can ingest the data. This allows the SIEM to correlate the APT event with other logs and incidents. It helps in threat hunting, compliance, and building a complete attack timeline.

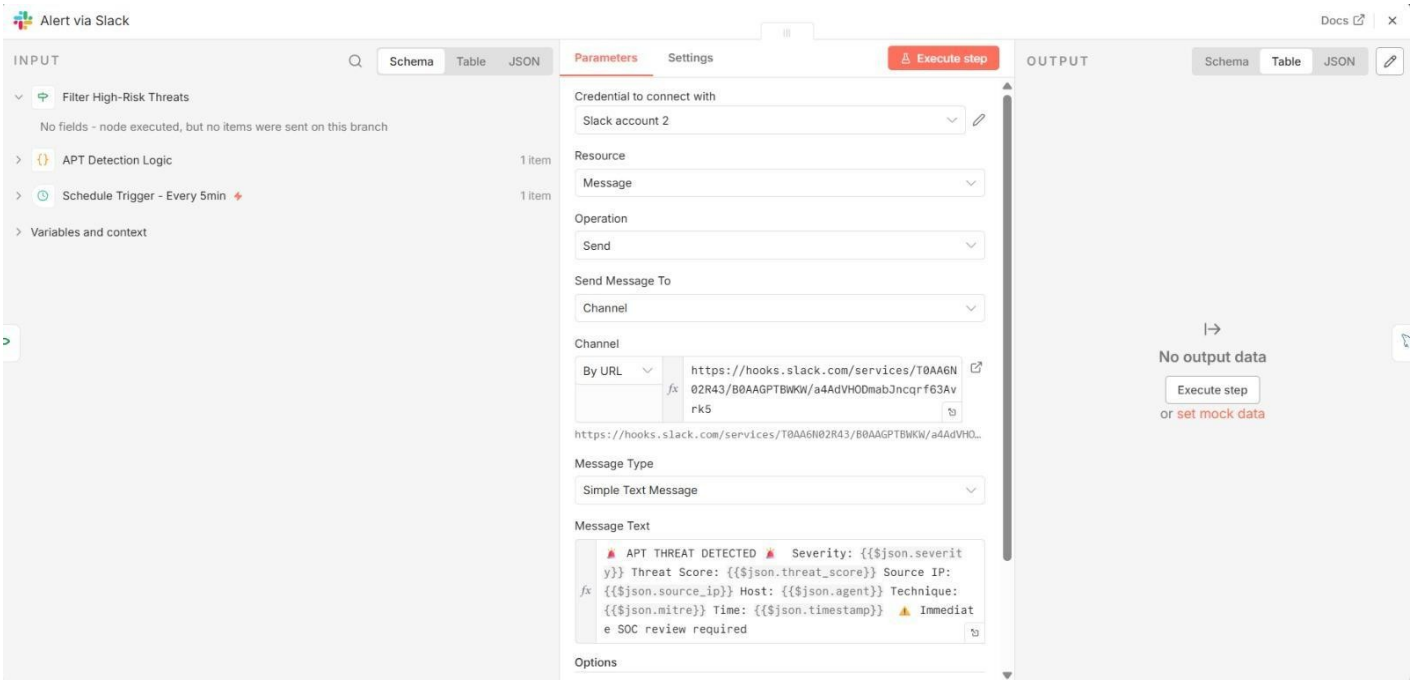


Fig 18: Alerts via Slack

This step shows real-time APT alerting through Slack using n8n automation. After Wazuh detects a high-risk APT event, only critical alerts pass through the filter and reach this node. n8n connects to Slack using a secure webhook and sends a formatted message containing severity, threat score, source IP, host, MITRE technique, and timestamp. This allows SOC analysts to see and react to threats instantly. It improves response speed and collaboration during active security incidents.

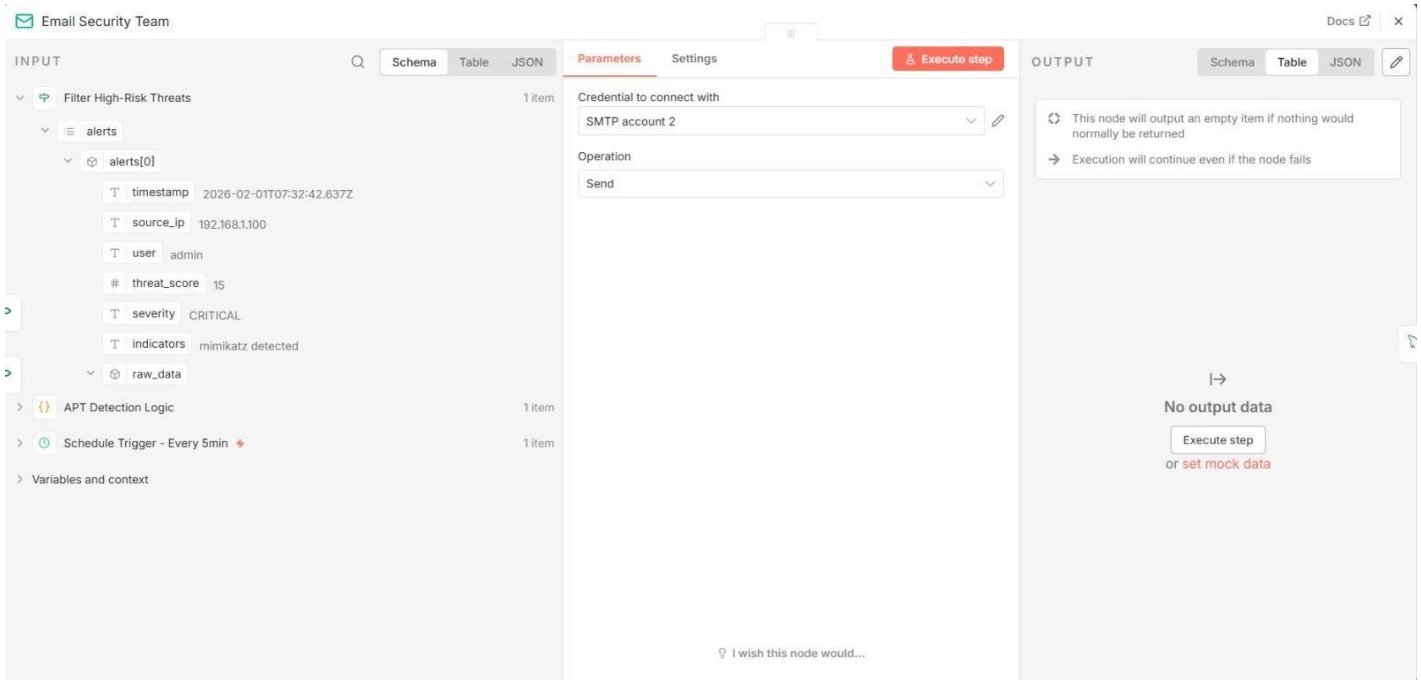


Fig 19: Email Security Team

This step shows how n8n notifies the security team by email when a high-risk APT alert is detected by Wazuh SIEM/EDR. After filtering only CRITICAL threats like mimikatz activity, the alert data (IP, user, severity, timestamp) is passed to this email node. n8n uses a configured SMTP account to automatically send the alert. This ensures the security team is informed in real time for investigation or escalation. It adds a human-in-the-loop layer to the automated APT response workflow.

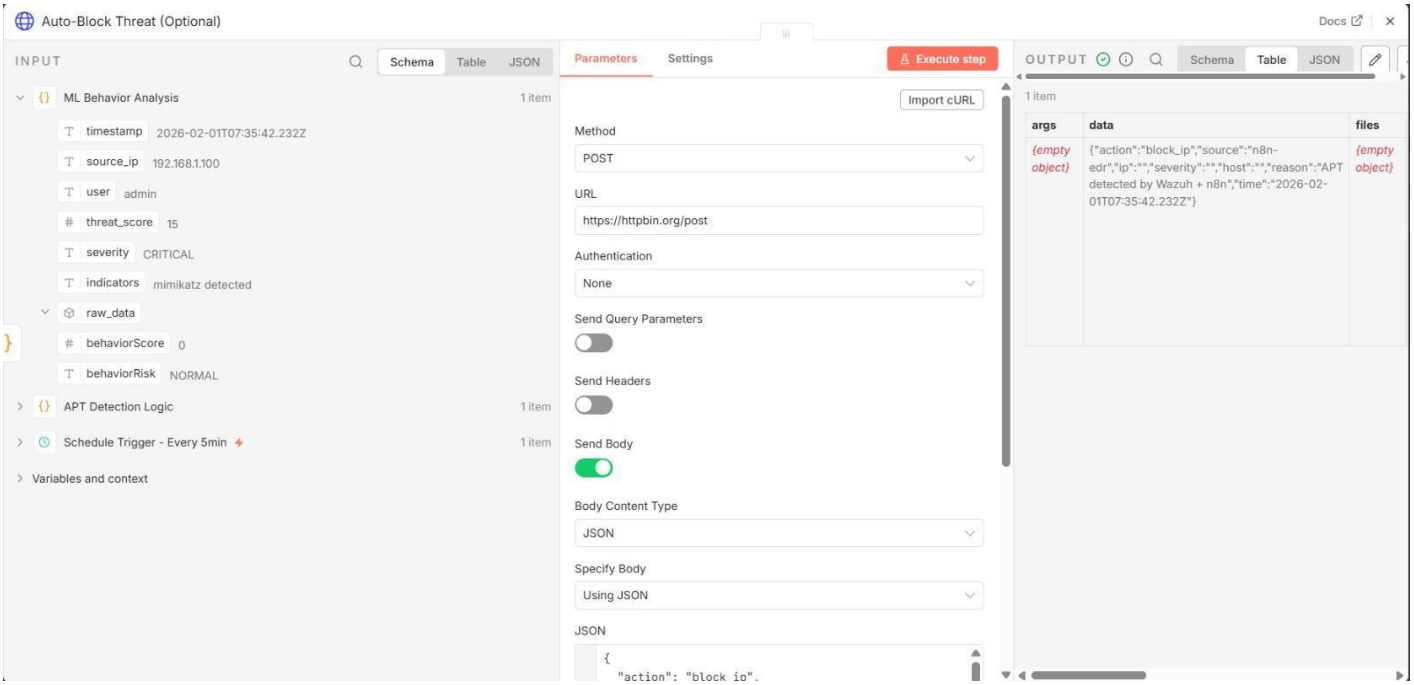


Fig 20: Auto Block threat

This structure shows an automated APT response where Wazuh EDR/SIEM detects advanced threats like credential dumping using behavior analysis and threat scoring. Once a critical APT alert is confirmed, the event is sent to n8n, which acts as the automation engine. n8n evaluates the alert logic and automatically triggers a response action. It then sends a JSON command via HTTP to block the attacker's IP or contain the threat. This setup reduces response time and limits attacker persistence in the network.

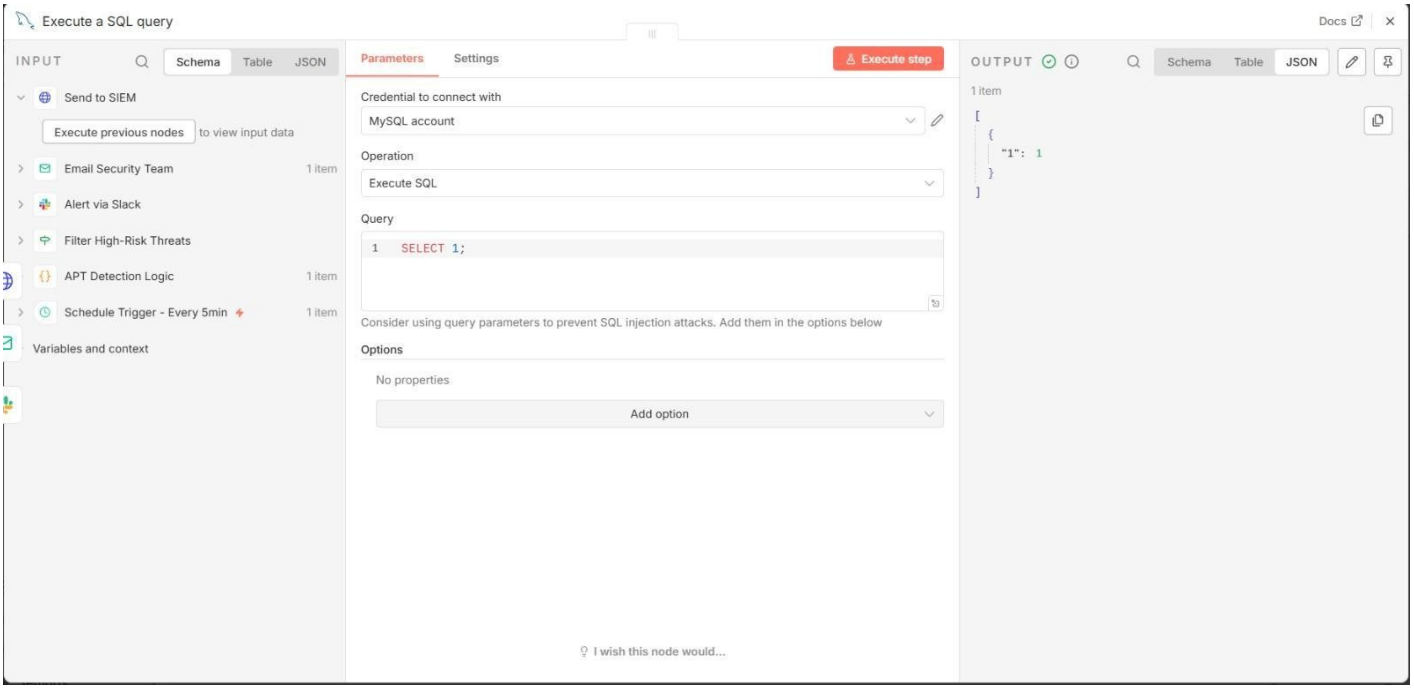


Fig 21: SQL Database

This step uses n8n to execute a SQL query against a connected database after an APT event is processed. It connects securely to a MySQL database using stored credentials. In a real APT workflow, this node would log incidents, update threat status, or store investigation data. The shown query is a test (SELECT 1) to confirm database connectivity. This helps maintain a historical record of security events for auditing and analysis.

13. Result:

13.1 Automated Email Alert for APT Detection

Fig 22: Email Alert

The figure illustrates the successful generation of an automated email alert triggered by the Advanced Persistent Threat (APT) detection pipeline implemented using Wazuh EDR integrated with n8n. When a high-risk security event is detected and classified as critical by the APT detection logic, the automation workflow immediately sends an alert email to the security team without any manual intervention.

The email clearly indicates that an APT Detection Alert has been generated, along with the associated risk level and the source of detection, identified as the *n8n APT Pipeline*. This ensures that the recipient can quickly understand the severity and origin of the alert. The message explicitly requests immediate investigation, emphasizing the critical nature of the detected activity.

This result demonstrates the effectiveness of the automated incident response mechanism. Instead of relying on manual monitoring of dashboards, security analysts are proactively notified in real time, enabling faster response and reduced dwell time for potential APT attacks. The alert also confirms the

successful integration of n8n with email services (SMTP) and validates the end-to-end workflow from detection to notification.

Overall, this outcome proves that the proposed system is capable of real-time alerting, rapid escalation, and reliable communication, which are essential requirements for mitigating Advanced Persistent Threats in enterprise environments.

13.2 n8n-Based APT Detection and Automated Response Workflow

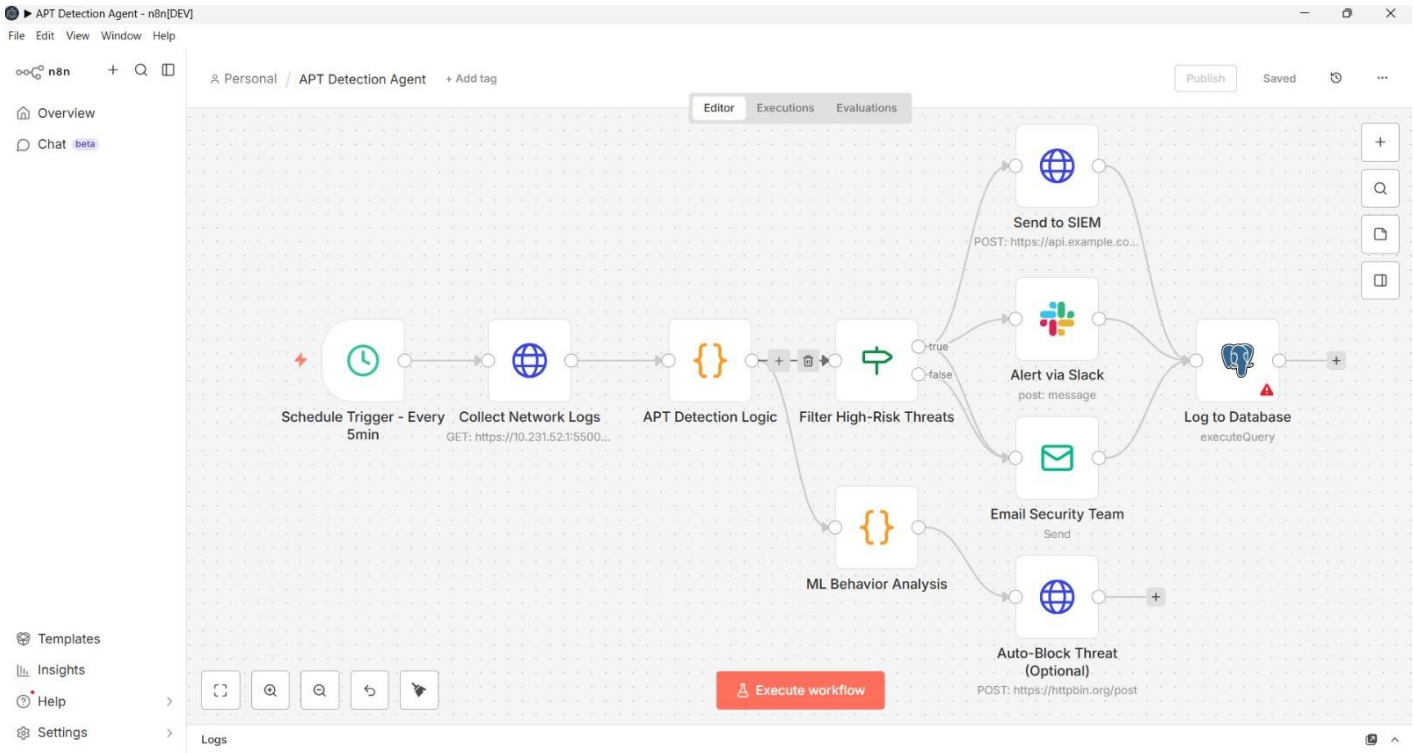


Fig 23: Executed Workflow

The figure demonstrates the successful execution of the n8n-based Advanced Persistent Threat (APT) Detection Agent developed in this project. The workflow represents a complete automated security pipeline that integrates log collection, threat analysis, alert classification, and incident response using Wazuh EDR and n8n automation.

The workflow begins with a Schedule Trigger configured to run every five minutes, ensuring continuous monitoring of security events without manual intervention. This periodic execution guarantees timely detection of suspicious activities associated with APT behavior. The trigger initiates the Collect Network

Logs stage, where n8n securely retrieves security alerts from the Wazuh Manager using its REST API.

The collected alerts are processed by the APT Detection Logic node, which applies correlation rules and assigns threat scores to identify potential APT indicators. This stage converts raw security data into structured alerts suitable for automated decision-making. The alerts are then evaluated by the Filter High-Risk Threats node, which classifies events based on severity and risk level. Only high-risk and critical alerts are allowed to proceed further, reducing alert noise and improving detection accuracy.

Once a high-risk threat is identified, the workflow triggers multiple automated response actions in parallel. Real-time notifications are sent to the security team through Slack, enabling immediate awareness of critical incidents. At the same time, email alerts are generated to ensure formal escalation and human intervention. The detected APT events are also forwarded to a central SIEM system for correlation, investigation, and long-term analysis. Additionally, all processed alerts are logged into a database to maintain an audit trail for compliance and forensic purposes.

The workflow includes an optional Machine Learning Behavior Analysis stage that enriches alerts with behavioral insights, helping detect stealthy or slow-moving APT activities. An Auto-Block Threat node is included as a future enhancement, demonstrating the system's capability to perform automated containment actions once detection accuracy is fully validated.

This result confirms that the proposed system successfully automates the entire APT detection and response lifecycle. The workflow reduces manual effort, improves response time, and provides a scalable and reliable solution for handling Advanced Persistent Threats in enterprise environments.

14. Conclusion :-

This project presented an effective approach for Advanced Persistent Threat (APT) detection and automated incident response using Wazuh EDR integrated with n8n workflow automation. The system enabled continuous monitoring of endpoints, detection of suspicious activities associated with APT techniques, and automated handling of security incidents. By combining Wazuh's detection and correlation capabilities with n8n's automation features, the proposed solution reduced manual intervention, improved response time, and enhanced overall security visibility. The results demonstrate that integrating EDR with automated response mechanisms is a practical and scalable solution for mitigating advanced cyber threats in modern enterprise environments.

Furthermore, the automated workflow helped in minimizing alert fatigue and ensured consistent incident handling across the infrastructure. The centralized logging and alert correlation improved forensic analysis and compliance reporting. The modular architecture allows easy enhancement and integration with additional security tools. Overall, the project highlights the importance of automation in strengthening modern cybersecurity operations.

15. Future Scope:-

The proposed system for Advanced Persistent Threat (APT) detection using Wazuh EDR and n8n can be further enhanced in several ways. Advanced machine learning and user behavior analytics can be integrated to improve detection of unknown and zero-day APT attacks. Automated containment actions such as endpoint isolation, IP blocking, and account disabling can be safely enabled with proper approval and rollback mechanisms.

The framework can be extended by integrating external threat intelligence feeds to enrich alerts with real-time Indicators of Compromise (IOCs). Support for cloud and hybrid environments, including container and cloud workload monitoring, can further increase its applicability. Additionally, integrating ticketing and case management systems would improve SOC workflow efficiency and make the solution suitable for large-scale enterprise deployments.

16. Reference :-

Table 5: References of the research paper & Its Authors

Ref No.	Paper Title	Authors	Journal / Conference	Year	Key Contribution	Relevance to Project
1	A Survey on Advanced Persistent Threat Detection Techniques	M. Ussath, D. Jaeger	IEEE Communications Surveys & Tutorials	2016	Survey of APT lifecycle and detection techniques	Defines APT lifecycle and threat behavior
2	Detection of Advanced Persistent Threats Using Log Correlation	J. François et al.	IEEE Cyber Security Conference	2015	Log correlation-based APT detection	Basis for log analysis using EDR
3	Endpoint Detection and Response: A Technical Review	S. Behl, A. Behl	Journal of Information Security	2017	EDR architecture and response mechanisms	Supports Wazuh EDR usage
4	Security Orchestration, Automation and	Gartner	Gartner Security Report	2019	SOAR automation workflows	n8n mapped as SOAR engine

	Response (SOAR): Trends					
5	Automated Cyber Defense Using Security Orchestration	K. R. Choo et al.	IEEE Access	2020	Automated incident response systems	Justifies automated response via n8n
6	Real-Time Intrusion Detection Using SIEM and EDR	A. Alshamrani et al.	Computers & Security	2019	SIEM + EDR integration	Wazuh SIEM-EDR foundation
7	Kill Chain-Based Analysis of Advanced Persistent Threats	E. Hutchins et al.	Lockheed Martin White Paper	2011	Cyber Kill Chain model	APT lifecycle modeling
8	Machine Learning Techniques for APT Detection	S. Zimba, Z. Wang	Future Generation Computer Systems	2021	ML-based APT detection	Behavioral detection concept
9	Incident Response Automation Using Open-Source Tools	P. Mahajan et al.	IJCN	2020	Open-source IR automation	Supports n8n usage

10	Design of Automated Security Incident Response Framework	R. Ahmad et al.	ACM Security Conference	2018	Automated IR framework	Overall system architecture reference